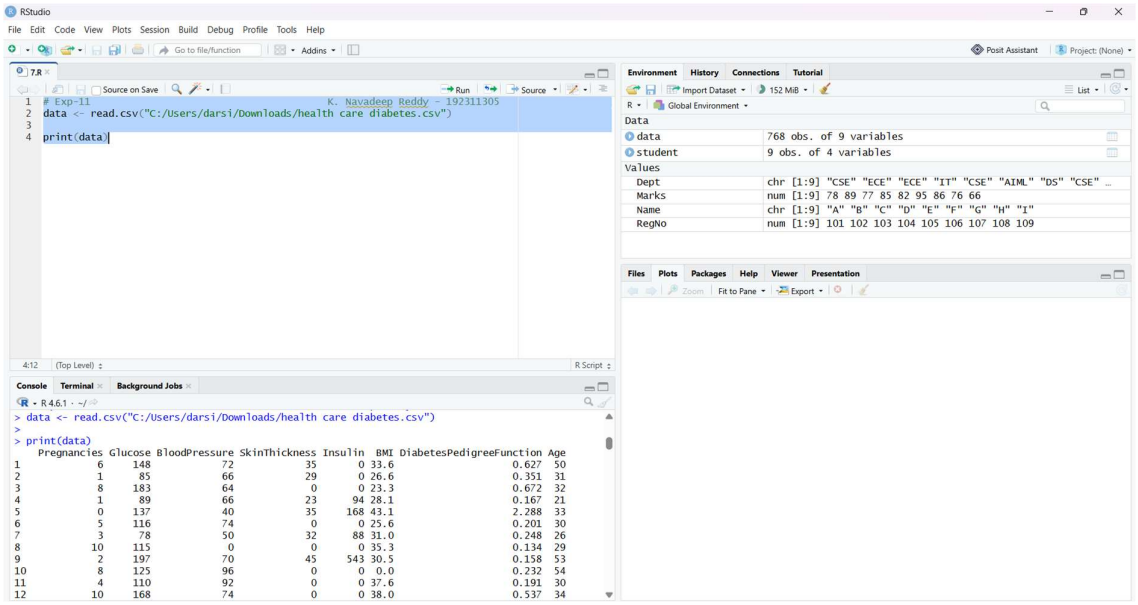
Experiments 11-20

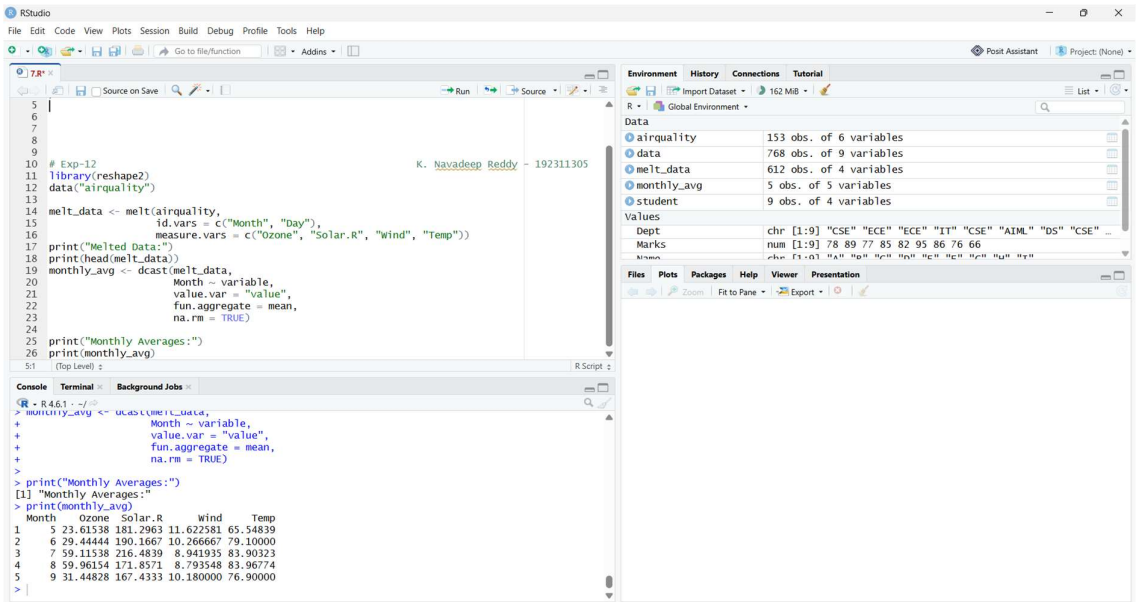
K. Navadeep Reddy

192311305

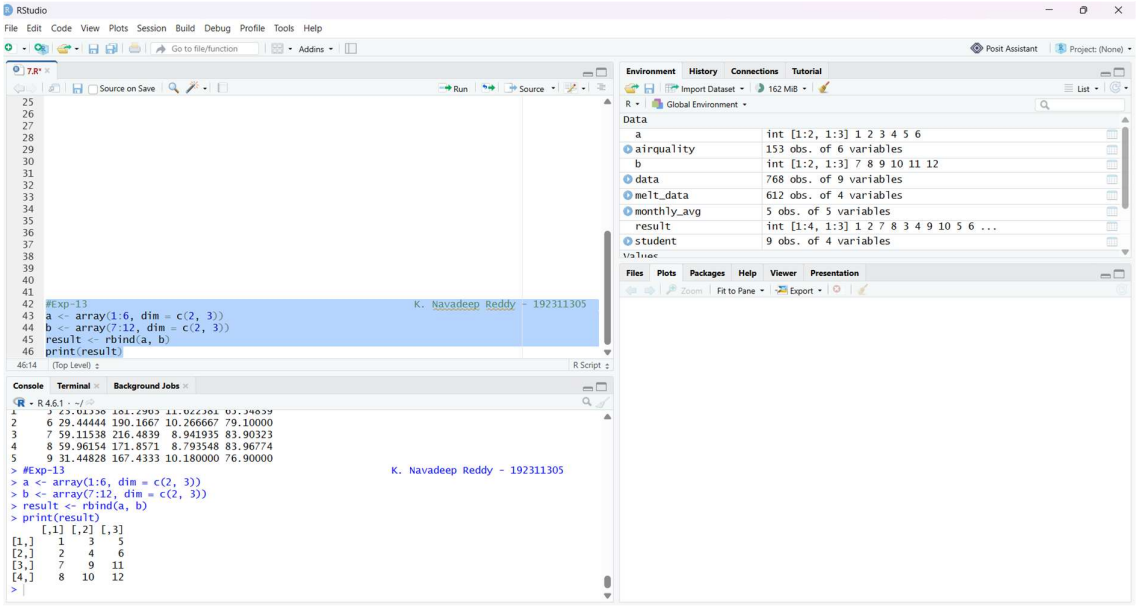
Exp-11



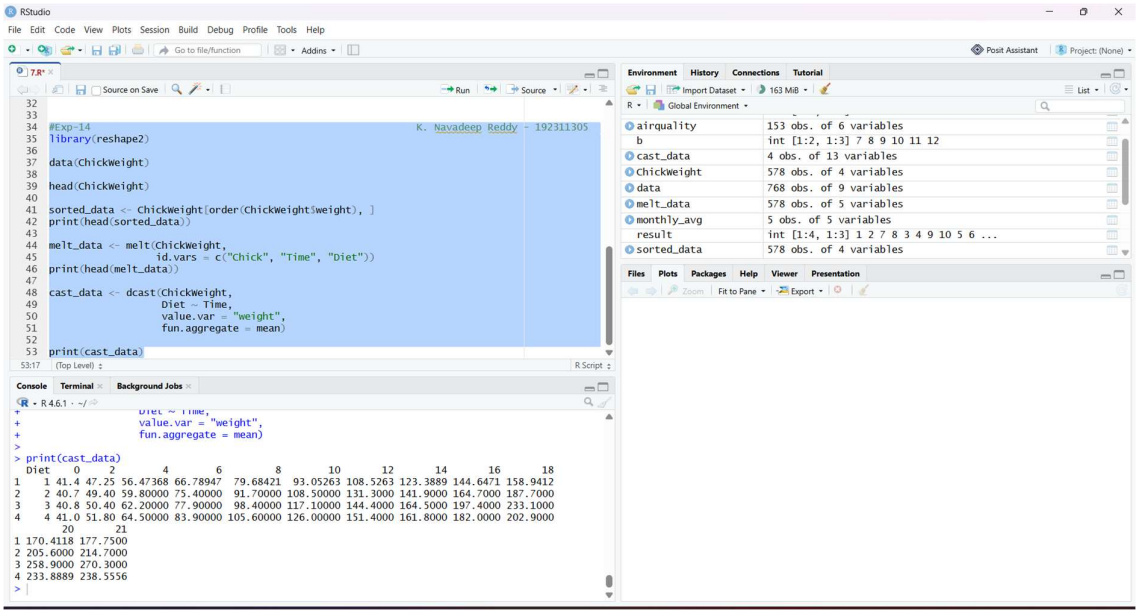
Exp-12



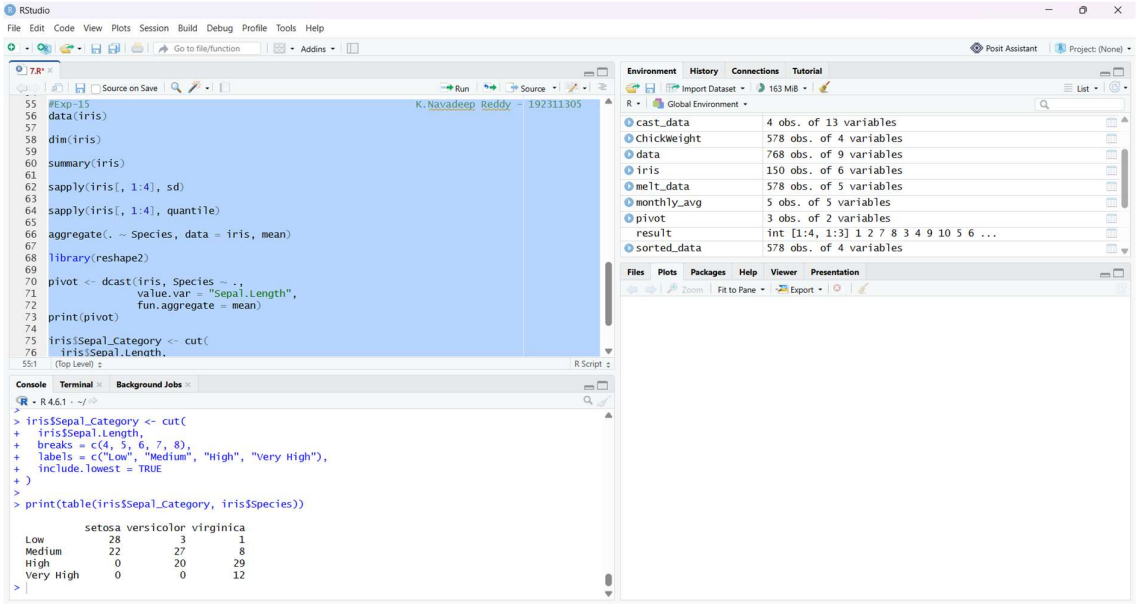
Exp-13


#Exp-13 K. Navadeep Reddy - 192311305
a <- array(1:6, dim = c(2, 3))
b <- array(7:12, dim = c(2, 3))
result <- rbind(a, b)
print(result)
> #Exp-13 K. Navadeep Reddy - 192311305
> a <- array(1:6, dim = c(2, 3))
> b <- array(7:12, dim = c(2, 3))
> result <- rbind(a, b)
> print(result)
[,1] [,2] [,3]
[1,] 1 3 5
[2,] 2 4 6
[3,] 7 9 11
[4,] 8 10 12

Exp-14


#Exp-14 K. Navadeep Reddy - 192311305
library(reshape2)
data(ChickWeight)
head(ChickWeight)
sorted_data <- ChickWeight[order(ChickWeight$weight),]
print(head(sorted_data))
melt_data <- melt(ChickWeight,
id.vars = c("Chick", "Time", "Diet"))
print(head(melt_data))
cast_data <- dcast(ChickWeight,
Diet ~ Time,
value.var = "weight",
fun.aggregate = mean)
print(cast_data)
> print(cast_data)
Diet 0 2 4 6 8 10 12 14 16 18
1 1 41.4 47.25 56.47368 66.78947 79.68421 93.05263 108.5263 123.3889 144.6471 158.9412
2 2 40.7 49.40 59.80000 75.40000 91.70000 108.50000 131.3000 141.9000 164.7000 187.7000
3 3 40.8 50.40 62.20000 77.90000 98.40000 117.10000 144.4000 164.5000 197.4000 233.1000
4 4 41.0 51.80 64.50000 83.90000 105.60000 126.00000 151.4000 161.8000 182.0000 202.9000
20 21
1 170.4118 177.7500
2 205.6000 214.7000
3 258.9000 270.3000
4 233.8889 238.5556

Exp-15



```
#Exp-15                                          K.Navadeep Reddy - 192311305
data(iris)

dim(iris)

summary(iris)

sapply(iris[, 1:4], sd)

sapply(iris[, 1:4], quantile)

aggregate(. ~ Species, data = iris, mean)

library(reshape2)

pivot <- dcast(iris, Species ~ .,
               value.var = "Sepal.Length",
               fun.aggregate = mean)
print(pivot)

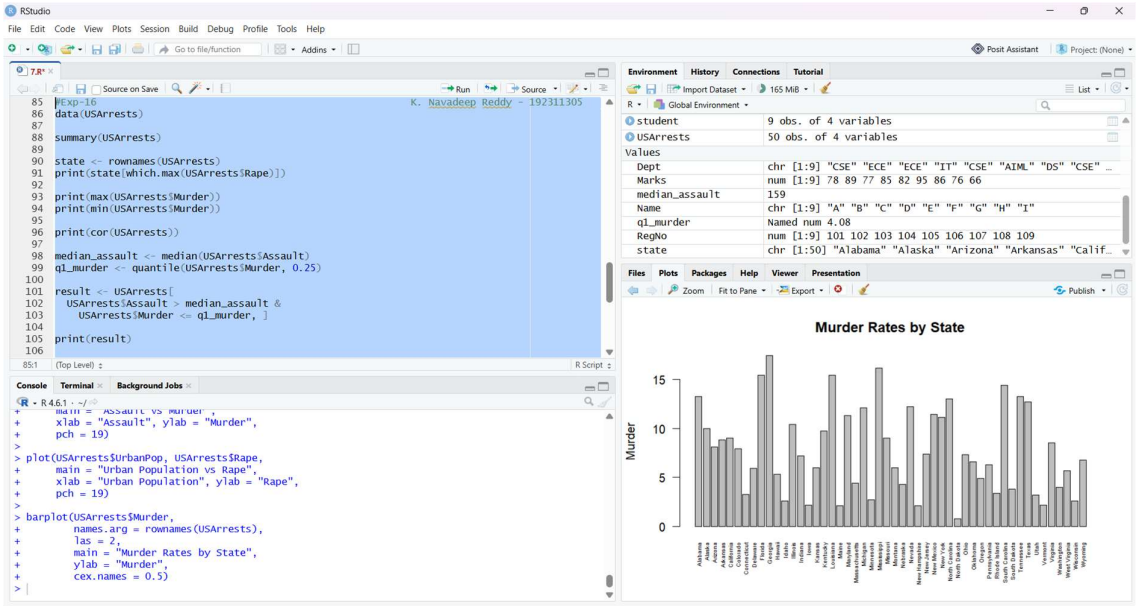
iris$Sepal_Category <- cut(
  iris$Sepal.Length,
  breaks = c(4, 5, 6, 7, 8),
  labels = c("Low", "Medium", "High", "Very High"),
  include.lowest = TRUE
)

print(table(iris$Sepal_Category, iris$Species))
```

```
> iris$Sepal_Category <- cut(
+   iris$Sepal.Length,
+   breaks = c(4, 5, 6, 7, 8),
+   labels = c("Low", "Medium", "High", "Very High"),
+   include.lowest = TRUE
+ )
>
> print(table(iris$Sepal_Category, iris$Species))

           setosa versicolor virginica
  Low          28          3         1
  Medium       22         27         8
  High          0         20        29
  Very High     0          0        12
>
```

Exp-16



```
#Exp-16                                          K. Navadeep Reddy - 192311305
data(USArrests)

summary(USArrests)

state <- rownames(USArrests)
print(state[which.max(USArrests$Rape)])

print(max(USArrests$Murder))
print(min(USArrests$Murder))

print(cor(USArrests))

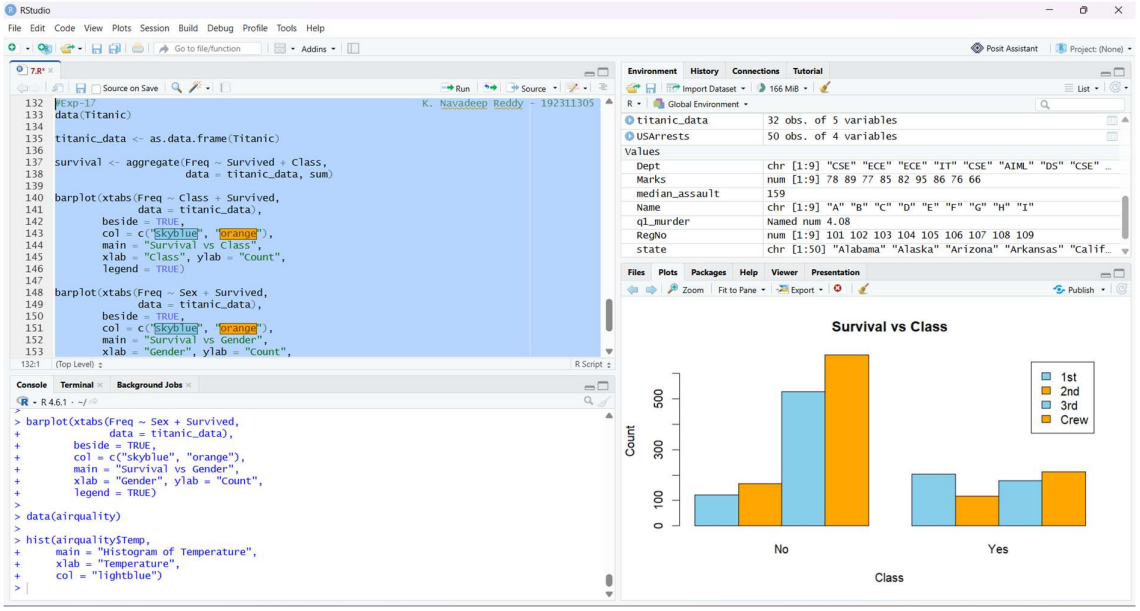
median_assault <- median(USArrests$Assault)
q1_murder <- quantile(USArrests$Murder, 0.25)

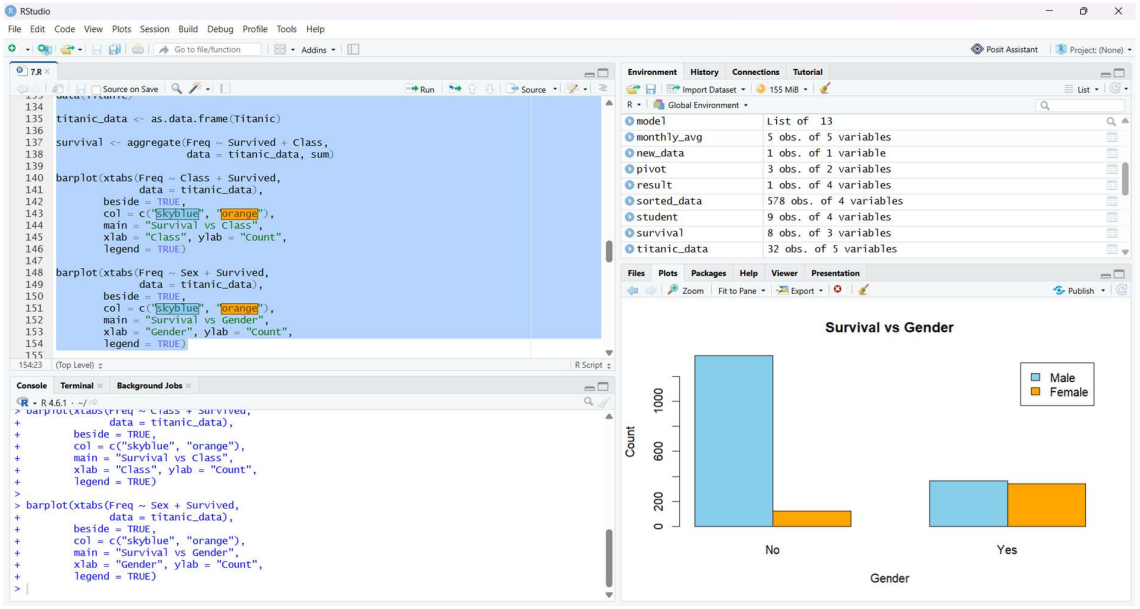
result <- USArrests[
  USArrests$Assault > median_assault &
    USArrests$Murder <= q1_murder, ]

print(result)
```

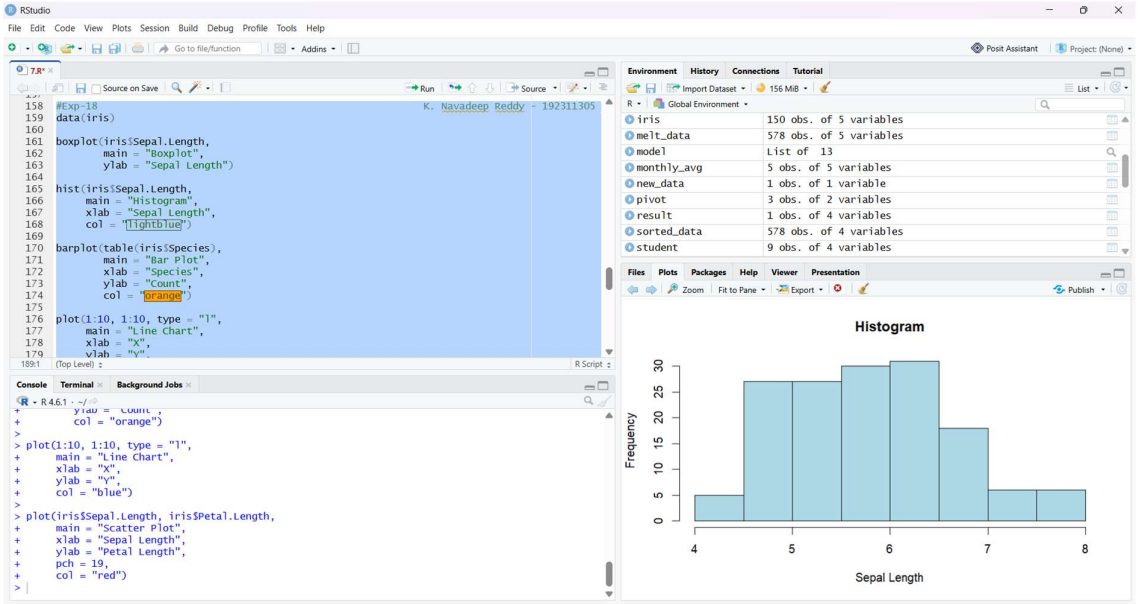
```
+       main = "Assault vs Murder",
+       xlab = "Assault", ylab = "Murder",
+       pch = 19)
>
> plot(USArrests$UrbanPop, USArrests$Rape,
+       main = "Urban Population vs Rape",
+       xlab = "Urban Population", ylab = "Rape",
+       pch = 19)
>
> barplot(USArrests$Murder,
+         names.arg = rownames(USArrests),
+         las = 2,
+         main = "Murder Rates by State",
+         ylab = "Murder",
+         cex.names = 0.5)
>
```

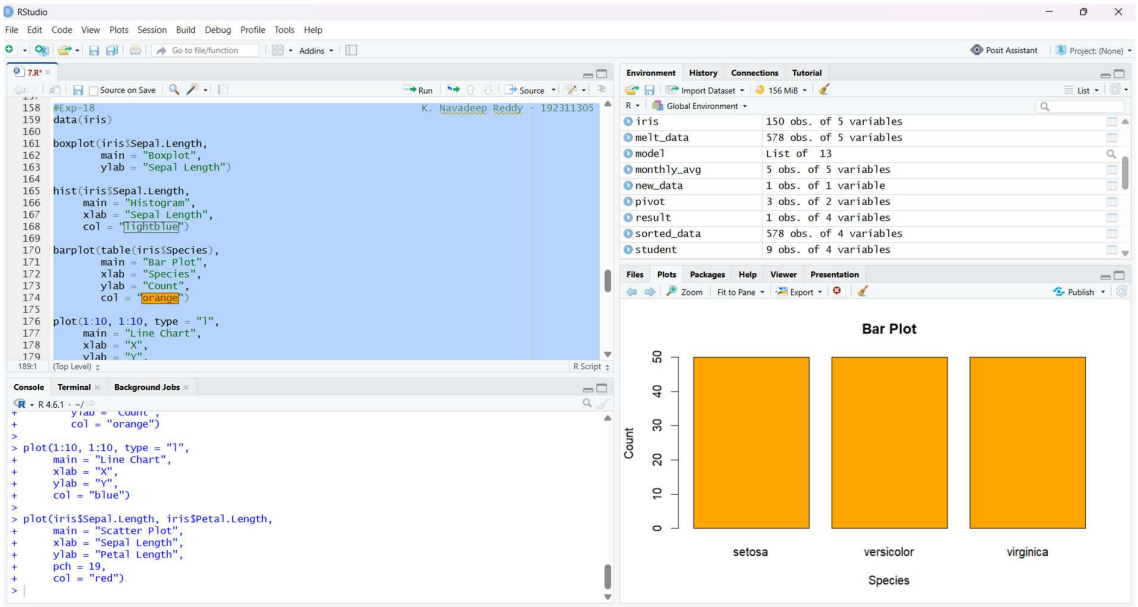
Exp-17

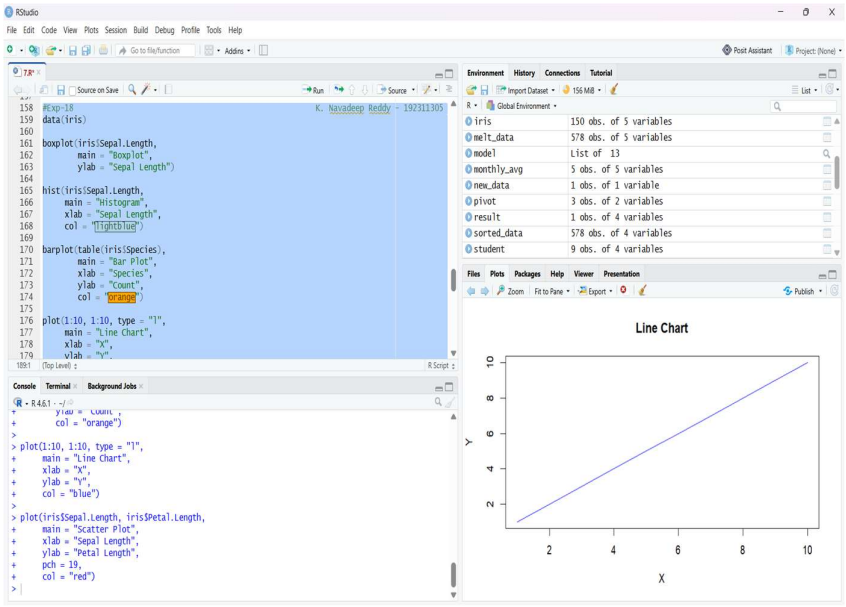

RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
#Exp-17 K. Navadeep Reddy - 192311305
data(Titanic)
titanic_data <- as.data.frame(Titanic)
survival <- aggregate(Freq ~ Survived + Class,
data = titanic_data, sum)
barplot(xtabs(Freq ~ Class + Survived,
data = titanic_data),
beside = TRUE,
col = c("skyblue", "orange"),
main = "Survival vs Class",
xlab = "Class", ylab = "Count",
legend = TRUE)
barplot(xtabs(Freq ~ Sex + Survived,
data = titanic_data),
beside = TRUE,
col = c("skyblue", "orange"),
main = "Survival vs Gender",
xlab = "Gender", ylab = "Count",
> barplot(xtabs(Freq ~ Sex + Survived,
+ data = titanic_data),
+ beside = TRUE,
+ col = c("skyblue", "orange"),
+ main = "Survival vs Gender",
+ xlab = "Gender", ylab = "Count",
+ legend = TRUE)
> data(airquality)
> hist(airquality$Temp,
+ main = "Histogram of Temperature",
+ xlab = "Temperature",
+ col = "lightblue")
titanic_data 32 obs. of 5 variables
USArrests 50 obs. of 4 variables
Values
Dept chr [1:9] "CSE" "ECE" "ECE" "IT" "CSE" "AIML" "DS" "CSE" ...
Marks num [1:9] 78 89 77 85 82 95 86 76 66
median_assault 159
Name chr [1:9] "A" "B" "C" "D" "E" "F" "G" "H" "I"
q1_murder Named num 4.08
RegNo num [1:9] 101 102 103 104 105 106 107 108 109
state chr [1:50] "Alabama" "Alaska" "Arizona" "Arkansas" "Calif...
Survival vs Class
Count
0
100
300
500
No
Yes
Class
1st
2nd
3rd
Crew

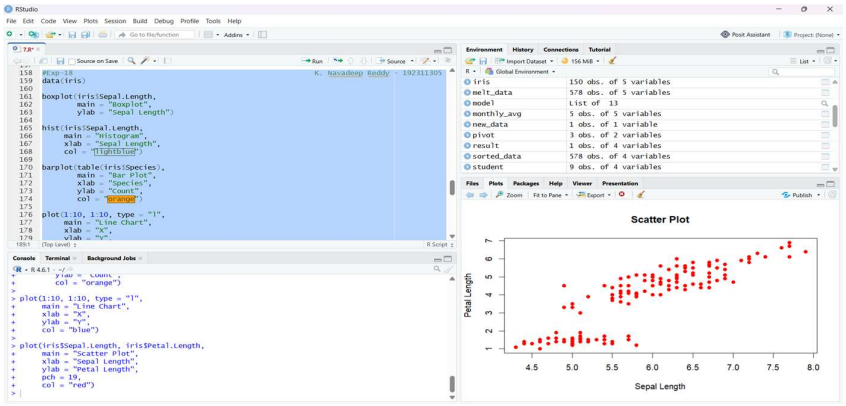

RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
titanic_data <- as.data.frame(Titanic)
survival <- aggregate(Freq ~ Survived + Class,
data = titanic_data, sum)
barplot(xtabs(Freq ~ Class + Survived,
data = titanic_data),
beside = TRUE,
col = c("skyblue", "orange"),
main = "Survival vs Class",
xlab = "Class", ylab = "Count",
legend = TRUE)
barplot(xtabs(Freq ~ Sex + Survived,
data = titanic_data),
beside = TRUE,
col = c("skyblue", "orange"),
main = "Survival vs Gender",
xlab = "Gender", ylab = "Count",
legend = TRUE)
> barplot(xtabs(Freq ~ Class + Survived,
+ data = titanic_data),
+ beside = TRUE,
+ col = c("skyblue", "orange"),
+ main = "Survival vs Class",
+ xlab = "Class", ylab = "Count",
+ legend = TRUE)
> barplot(xtabs(Freq ~ Sex + Survived,
+ data = titanic_data),
+ beside = TRUE,
+ col = c("skyblue", "orange"),
+ main = "Survival vs Gender",
+ xlab = "Gender", ylab = "Count",
+ legend = TRUE)
model List of 13
monthly_avg 5 obs. of 5 variables
new_data 1 obs. of 1 variable
pivot 3 obs. of 2 variables
result 1 obs. of 4 variables
sorted_data 578 obs. of 4 variables
student 9 obs. of 4 variables
survival 8 obs. of 3 variables
titanic_data 32 obs. of 5 variables
Survival vs Gender
Count
0
200
600
1000
No
Yes
Gender
Male
Female

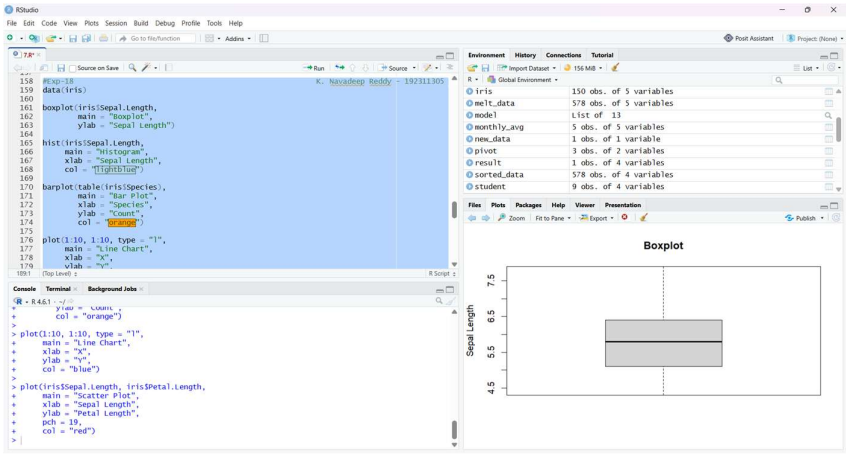
Exp-18


#Exp-18 K. Navadeep Reddy - 192311305
data(iris)
boxplot(iris$Sepal.Length,
main = "Boxplot",
ylab = "Sepal Length")
hist(iris$Sepal.Length,
main = "Histogram",
xlab = "Sepal Length",
col = "lightblue")
barplot(table(iris$Species),
main = "Bar Plot",
xlab = "Species",
ylab = "Count",
col = "orange")
plot(1:10, 1:10, type = "l",
main = "Line Chart",
xlab = "X",
ylab = "Y",
col = "blue")
plot(iris$Sepal.Length, iris$Petal.Length,
main = "Scatter Plot",
xlab = "Sepal Length",
ylab = "Petal Length",
pch = 19,
col = "red")
Histogram
Frequency
Sepal Length

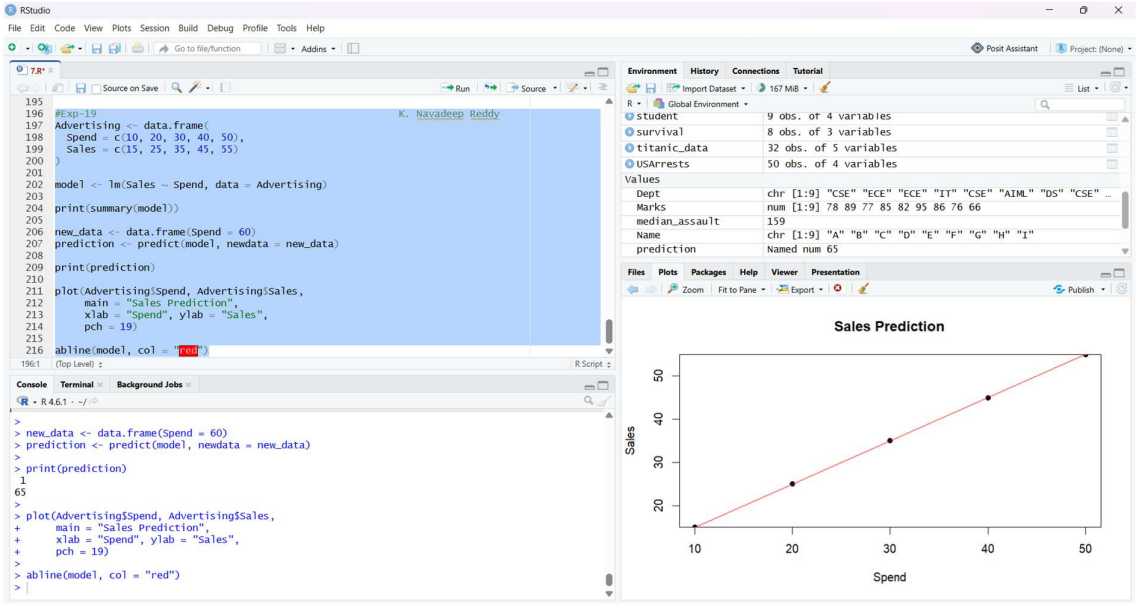

#Exp-18 K. Navadeep Reddy - 192311305
data(iris)
boxplot(iris$Sepal.Length,
main = "Boxplot",
ylab = "Sepal Length")
hist(iris$Sepal.Length,
main = "Histogram",
xlab = "Sepal Length",
col = "lightblue")
barplot(table(iris$Species),
main = "Bar Plot",
xlab = "Species",
ylab = "Count",
col = "orange")
plot(1:10, 1:10, type = "l",
main = "Line Chart",
xlab = "X",
ylab = "Y",
col = "blue")
plot(iris$Sepal.Length, iris$Petal.Length,
main = "Scatter Plot",
xlab = "Sepal Length",
ylab = "Petal Length",
pch = 19,
col = "red")
Bar Plot
Count
setosa
versicolor
virginica
Species


#Exp-18 K. Navadeep Reddy - 192311305
data(iris)
boxplot(iris$Sepal.Length,
main = "Boxplot",
ylab = "Sepal Length")
hist(iris$Sepal.Length,
main = "Histogram",
xlab = "Sepal Length",
col = "lightblue")
barplot(table(iris$Species),
main = "Bar Plot",
xlab = "Species",
ylab = "Count",
col = "orange")
plot(1:10, 1:10, type = "l",
main = "Line Chart",
xlab = "X",
ylab = "Y",
col = "blue")
plot(iris$Sepal.Length, iris$Petal.Length,
main = "Scatter Plot",
xlab = "Sepal Length",
ylab = "Petal Length",
pch = 19,
col = "red")
Line Chart
Y
X

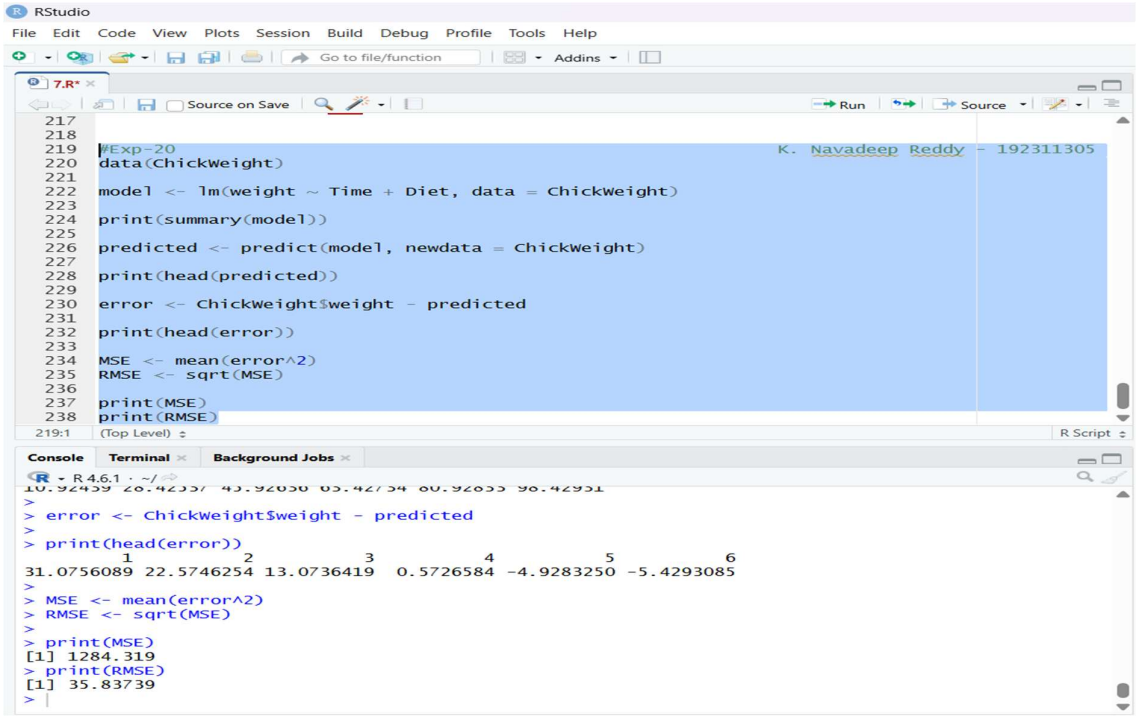

Scatter Plot
Petal Length
Sepal Length


Boxplot
Sepal Length

Exp-19


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
#Exp-19 K. Navadeep Reddy
Advertising <- data.frame(
Spend = c(10, 20, 30, 40, 50),
Sales = c(15, 25, 35, 45, 55)
)
model <- lm(Sales ~ Spend, data = Advertising)
print(summary(model))
new_data <- data.frame(Spend = 60)
prediction <- predict(model, newdata = new_data)
print(prediction)
plot(Advertising$Spend, Advertising$Sales,
main = "Sales Prediction",
xlab = "Spend", ylab = "Sales",
pch = 19)
abline(model, col = "red")
Console Terminal Background Jobs
> new_data <- data.frame(Spend = 60)
> prediction <- predict(model, newdata = new_data)
> print(prediction)
1
65
> plot(Advertising$Spend, Advertising$Sales,
+ main = "Sales Prediction",
+ xlab = "Spend", ylab = "Sales",
+ pch = 19)
> abline(model, col = "red")
Environment History Connections Tutorial
student 9 obs. of 4 variables
survival 8 obs. of 3 variables
titanic_data 32 obs. of 5 variables
USArrests 50 obs. of 4 variables
Values
Dept chr [1:9] "CSE" "ECE" "ECE" "IT" "CSE" "AIML" "DS" "CSE" ...
Marks num [1:9] 78 89 77 85 82 95 86 76 66
median_assault 159
Name chr [1:9] "A" "B" "C" "D" "E" "F" "G" "H" "I"
prediction Named num 65
Files Plots Packages Help Viewer Presentation
Sales Prediction
Sales
Spend
10 20 30 40 50
20 30 40 50

Exp-20


RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
#Exp-20 K. Navadeep Reddy - 192311305
data(ChickWeight)
model <- lm(weight ~ Time + Diet, data = ChickWeight)
print(summary(model))
predicted <- predict(model, newdata = ChickWeight)
print(head(predicted))
error <- ChickWeight$weight - predicted
print(head(error))
MSE <- mean(error^2)
RMSE <- sqrt(MSE)
print(MSE)
print(RMSE)
Console Terminal Background Jobs
> error <- ChickWeight$weight - predicted
> print(head(error))
1 2 3 4 5 6
31.0756089 22.5746254 13.0736419 0.5726584 -4.9283250 -5.4293085
> MSE <- mean(error^2)
> RMSE <- sqrt(MSE)
> print(MSE)
[1] 1284.319
> print(RMSE)
[1] 35.83739